Chapter 3 Notes

3.1

In the proof of Theorem 3.1, here is why the class of all transitive natural numbers is a set. We can show that it is a subset of ω, and is hence a set by A_2. For suppose x is a member of the class of all transitive natural numbers. Then obviously x is also a natural number, hence $x \in \omega$, thus completing the proof.

Note that here is why the only natural numbers allowed for by Definition 2.2 are the ones defined right before Definition 1.1. By Definition 2.2, a natural number is a member of every inductive set. Now, consider the following inductive set: $\emptyset$ is an element, and if x is an element, then so is x^+. Hence every natural number is a member of this set. But this set is just the set of all sets that were defined in the way just before Definition 1.1.

Exercise 3.1

Suppose that A is a class of natural numbers such that for every natural number n, if A contains all numbers less than n, then A contains n as well. Now, we can prove by induction that A contains all natural numbers. Consider the property P meaning A contains all numbers natural numbers less than or equal to n. Now, there are no natural numbers less than 0. Thus the statement that A contains all natural numbers less than 0 is vacuously true; hence A contains 0. Thus, because 0 is the only natural number less than or equal to 0, P holds for 0. Now suppose that P holds for some n, that is, A contains all natural numbers less than or equal to n. But then A contains all numbers less than $n+1$, meaning (by the definition of A) A also contains $n+1$. Thus, by induction, A contains all natural numbers.

Exercise 3.2

Consider a natural number n.

(a): Suppose that $x \in n$. Now, it cannot be that $n = \emptyset$. Thus n is non-empty, hence of the form $(n-1)^+$. Thus either $x \in n-1$ or $x = n-1$. Now, $\cup n$ is just the set of all elements of all elements of $n-1$ together with all elements of $n-1$. Now suppose $x = n-1$. Then $x \notin \cup n$. For suppose it were. Then it would either be an element of $n-1$ or it would be an element of an element of $n-1$. In the first case, we would have $n-1 \in n-1$, which is ruled out by Theorem 3.2. In the second case, there would be some y such that $x \in y$ and $y \in n-1$. But then, by the transitivity of $n-1$ (by Theorem 3.1), $x \in n-1$, that is, $n-1 \in n-1$, which is again ruled out by Theorem 3.2. Thus $n-1 \notin \cup n$, meaning the statement is not necessarily true.

It should be noted that all elements other than $n-1$ of n are members of $\cup n$. For all elements of n

are either elements of $n-1$ or equal to $n-1$. Thus, if they are not equal to $n-1$, the element is an element of $n-1$. But, because $n-1 \in n$, then all elements of $n-1$ are elements of $\cup n$.

(b): Suppose that $x \in \cup n$. Thus either $x \in n-1$ or x is an element of some element of $n-1$. In the first case, because $n-1 \in n$, $x \in n$ (by the transitivity of n). In the second case, because $n-1$ is transitive, it follows that $x \in n-1$. But then, because $n-1 \in n$ and n is transitive, $x \in n$. Thus, in either case, $x \in n$, and thus $\cup n \subseteq n$. Hence the statement is necessarily true.

(c): By part (b) of this problem, it follows that $x \in \cup n$ implies $x \in n$. But $n^+ = n \cup \{n\}$. Thus all elements of n are in n^+, meaning $x \in n^+$. On the other hand, suppose that $x \in n^+$. Then it could be that $x = n$. But then, if $x \in \cup n$, we would have either n being an element of an element of $n-1$, or $n \in n-1$. In the first case, the transitivity of $n-1$ would show that $n \in n-1$. Thus we would have, in either case, $n \in n-1$, which is impossible. For if $n \in n-1$, then, because $n-1 \in n$, $n-1 \in n-1$ by the transitivity of $n-1$, which is impossible (or, if $n-1 = \emptyset$, it could have no members). Hence it is not necessarily the case that $n^+ \subseteq \cup n$. Thus $\cup n = n^+$ does not necessarily hold.

Similar reasoning as to the above also shows that $n-1$ cannot be an element of $\cup n$. It should be noted that $n-1$ and n are the only members of n^+ that are not members of $\cup n$. For any other element of n^+ is an element of n but not equal to $n-1$. But $n = n-1 \cup \{n-1\}$. Hence it would be an element of $n-1$, and because $n-1$ is an element of n, it would also be an element of $\cup n$.

(d): On the one hand, suppose that $x \in \cup n^+$. Then either x is an element of an element of n, or x is an element of n. In the first case, because n is transitive, $x \in n$. The second case means $x \in n$. Thus, in any case, $x \in n$. On the other hand, suppose that $x \in n$. Then $x \in \cup n^+$, as $n^+ = n \cup \{n\}$. Hence $\cup n^+ = n$.

(e): By part (b) of this problem, $\cup n \subseteq n$ holds. But by the answer to part (a) of this problem, it does not necessarily follow that $n \subseteq \cup n$. Thus it is not necessarily true that $\cup n = n$.

Exercise 3.3

(a): On the one hand, suppose that $x \in \cup\omega$. Then x is an element of a natural number. Thus either $x \in n$ or $x = n$. Now we prove the following statement: every element of every natural number is a natural number. The base case vacuously holds, as $\emptyset$ has no elements. Now suppose the statement holds for some n. Then $n^+ = n \cup \{n\}$. Hence, if $y \in n^+$, either $y \in n$ or $y = n$. In the former case, by the induction assumption, it follows that y is a natural number, and in the latter case, y would be a natural number (as n is a natural number). Thus, in any case, y is a natural number, therefore completing the proof. As a consequence of having proven this statement, it follows that x is a natural number, and hence $x \in \omega$. On the other hand, suppose that $x \in \omega$. Thus, for some n, $x = n$. Now, all elements of n^+ are in $\cup\omega$. But n is an element of n^+. Thus $n \in \cup\omega$, and hence $x \in \cup\omega$. Thus we have proven that $\cup\omega = \omega$.

(b): On the one hand, suppose that $x \in \cup\omega^+$. Now, $\omega^+ = \omega \cup \{\omega\}$. Thus either x is an element of a natural number or x is a natural number. By the results of part (a) of this problem, it follows that if x is an element of a natural number, then x is a natural number. Thus x is a natural number, that is, $x \in \omega$. Now suppose that $x \in \omega$. Then $x \in \cup\omega^+$, as x would be an element of ω, which is an element of ω^+. Thus we have proven that $\cup\omega^+ = \omega$.

Note that a generalization cannot be made of this problem. Define $\{\omega_0, \omega_1, \omega_2, ..., \omega_n, ...\}$ in the following way: $\omega_0 = \omega$ and $\omega_{n+1} = \omega_n^+$. Then it does not follow that $\cup\omega_n = \omega$ for all n. For consider $n = 2$. This is ω_1^+, that is,

$$\omega_1 \cup \{\omega_1\} = (\omega \cup \{\omega\}) \cup \{(\omega \cup \{\omega\})\}$$

Thus $\omega \in \cup\omega_1^+$. But $\omega \notin \omega$, meaning $\cup\omega_2 \neq \omega$. Indeed, it is a general fact that, for $n \geq 2$, $\cup\omega_n \neq \omega$. To prove this, we can just prove that, for all $n \geq 2$, $\omega \in \omega_n$. We have just proven the base case. If the result

holds for some n, then $\omega \in \omega_{n+1}$, as $\omega_{n+1} = \omega_n \cup \{\omega_n\}$, and it was assumed that $\omega \in \omega_n$. Thus, for every $n \geq 2$, $\omega \in \omega_n$. Now we can prove the statement that, for $n \geq 2$, $\cup\omega_n \neq \omega$. We proved the base case for $n = 2$ above. Now consider some $n > 2$. Then $\omega_n = \omega_{n-1} \cup \{\omega_{n-1}\}$ and $\cup\omega_n$ consists of all elements of all elements of ω_{n-1} and all elements of ω_{n-1}. Because ω is an element of ω_{n-1}, it follows that $\omega \in \cup\omega_n$, and thus, because $\omega \notin \omega$, $\cup\omega_n \neq \omega$.

Section 4

Notes

Note that g-induction only works on minimally inductive sets for the following reason: If A was not minimally inductive under g, we would have some element x of A such that x is not a term of the g-sequence (see the section below titled Exercise 4.1) $\emptyset, g(\emptyset), g(g(\emptyset)), ...$ But the proof by g-induction only proves the result for the terms of the g-sequence started by $\emptyset$, meaning we would not necessarily know that the result held for x.

Note that (1) of Lemma 4.6 holds because 0 is just $\emptyset$, and $\emptyset$ obviously is a subclass of y for every y.

Theorem 4.16

The proof of Theorem 4.16 seems incorrect to me. At a point in the proof, we rely upon Theorem 4.12. But that assumes L is a bounded *subset* of A. However he assumes that $\emptyset \notin A$ and shows that $\emptyset \in L$, meaning L is not a subset of A, contrary to the claim that it was. (In fact, I think we would need to show that L is a subset of M, as M is the one which is minimally inductive under g, but Smullyan does not make this clear)

I think that what Smullyan was trying to say was: Let L be the class of all elements of x *of* M such that x is a proper subclass of all elements of A. Obviously, this means that L is a subset of M, and then, when we show that L is bounded by A, we really can use Theorem 4.12. Also, when we assume that $\emptyset \notin A$, we can still conclude that $\emptyset \in L$, as $\emptyset \in M$ by the assumption that M is minimally inductive under g.

Exercise 4.1

We suppose that N is closed under g, where g is progressing, and that b is an element of N and that no proper sublcass of N containing b is closed under g. In order to prove the desired results, we can first prove (1) of Theorem 4.17. We are given that N is closed under g. Call a sequence of the form $x, g(x), g(g(x)), g(g(g(x))), ...$ a $g - sequence$ (and call it a subset of a set A if all elements of the sequence are elements of A). Label x the first term of the g-sequence, $g(x)$ the second term, and, in general, if y is the nth term of the g-sequence, then $g(y)$ is the $n+1$th term of the g-sequence (and we say that n is the *index* of the nth term of the sequence). We first show that N is just the g-sequence $b, g(b), g(g(b)), ...$ Now, it is obvious that the g-sequence started by b is a subset of N (because N is closed under g, and b is in N). Then suppose that there was some element $c \in N$ outside of the g-sequence started by b. Then the g-sequence $c, g(x), g(g(c)), ...$ would be a subset of A. But then $b, g(b), g(g(b)), ...$ would be a proper subset of N containing b which was closed under g, contrary to the hypothesis. Hence the only elements in N are the elements of the g-sequence started by b. Having shown this, recall that g is a progressive function. Now consider any x, y in N. If $x = y$, then $y \subseteq x$. Now suppose $x \neq y$. Let x be the mth term of the g-sequence and y the nth term. Thus $m \neq n$. On the one hand, suppose that $m < n$. An obvious induction argument

shows that $x \subseteq y$. Now either $g(x) = y$ or not. In either case, $g(x) \subseteq y$ is obvious. On the other hand, suppose that $n < m$. Then the same argument shows that $y \subseteq x$, thus completing the proof of (1).

Here is the "obvious inductive argument" mentioned above (I will refer to this as Lemma 1). We prove that, for all n, all mth terms where $m < n$ are subsets of the nth term of the g-sequence. The base case is obvious (as there are no terms of index less than 1). Now suppose the result holds for the nth term x. Then all terms of index m where $m < n$ are subsets of x. But then, because g is progressing, x is a subset of $g(x)$ (the $n+1$th term), and because all the terms of index m where $m < n$ are subsets of x, it follows that all terms of index m where $m < n+1$ are subsets of $g(x)$, completing the proof.

We now prove (2) of the problem. We prove by induction on n that every non-empty subset of N that is bounded by the nth term of N (that is, the nth term of the g-sequence $b, g(b), g(g(b)), ...$) has a greatest element. The base case is when $n = 0$, that is, when the term is b. Now consider some bounded subset c of N which is bounded by b. Thus $c \subseteq N$ and, for all $d \in c$, $d \subseteq b$. But, by Lemma 1, b is a subset of all elements of N. Thus b is a subset of all elements $d \in c$. But $d \subseteq b$. Hence $b = d$ for all elements $d \in c$, meaning c is just the set $\{b\}$. Then, of course, b is the greatest element of c, therefore proving the base case. Now suppose the hypothesis holds for some n. That is, every non-empty subset of N bounded by the nth term x of N has a greatest element. Now, the $n+1$th term of N is just $g(x)$. Consider any non-empty subset B of N bounded by $g(x)$. If $g(x) \in B$, then of course $g(x)$ is the greatest element of B. So suppose that $g(x) \notin B$. Then every element of B is a proper subset of $g(x)$. Consider any element y of B. Then $y \subset g(x)$, meaning $g(x) \subseteq y$ doesn't hold, and therefore $y \subseteq x$ holds (because either $g(x) \subseteq y$ or $y \subseteq x$ by (1) of this problem). But this means that B is bounded by x; hence, by the induction assumption, it has a greatest element.

I was not entirely sure what (3) was trying to prove: I interpreted it to mean that, if x is the greatest element of N, then it is a fixed point of g. Suppose that x is the greatest element of N. Then, for all elements y of N, $y \subseteq x$. Now, we know that $g(x) \in N$. Because x is the greatest element of N, $g(x) \subseteq x$. But also, because g is progressing, $x \subseteq g(x)$. Thus $g(x) = x$, meaning x is a fixed point of g.

Now we prove (4) of the problem. Because we have shown that $g(x) \subseteq y$ or $y \subseteq x$ for all x, y in N, it follows that $x \subseteq y$ or $y \subseteq x$ (as $x \subseteq g(x)$). Thus N is a nest. It remains to be shown that every non-empty subclass of N has a least element. So consider some non-empty subclass A of N. Because N is just the set of all terms of the g-sequence $b, g(b), g(g(b)), ...$, it follows that A contains only members of that sequence. Hence A contains some term x whose index (in the g-sequence) n is less than or equal to the indices of all the other elements of A. By Lemma 1, it follows that x is a subset of all terms with an index greater than or equal to n, and hence x is a subset of all other elements of A, meaning it is the least element of A.

Because b starts the g-sequence $b, g(b), g(g(b)), ...$, which makes up the entirety of N, Lemma 1 above shows that b is the least element of N (is a subset of all elements of N). This proves (5) of the problem.

The reason that this generalizes some of the earlier results of the section is that it takes away the requirenment that N be minimally inductive under g. For example, Theorem 4.7 holds now when N is closed under g, where g is progressing, and that b is an element of N and that no propert subclass of N containing b is closed. That is, we do not need to know that N is minimally inductive under g.

Exericse 4.2

(I did not think that this proof requires A to be a set)
Suppose that A is inductive under g and that A is a set. Thus we know
(1) $\emptyset \in A$,
(2) for every x in A, the set $g(x)$ is in A.
Consider the g-sequence $\emptyset, g(\emptyset), g(g(\emptyset)), ...$ Because A is inductive under g, that g-sequence is a subset of

A. But clearly, that g-sequence is minimally inductive under g. Hence there is some subset of A that is minimally inductive under g.

While we cannot show that every g-inductive class is a set (as Smullyan remarked, our axioms are not yet powerful enough to show that every denumerable class is a set), we can show that every element of every g-inductive class is a set. Recall that the definition of a function relied upon the definition of relations, which in turn relied upon the notion of an ordered pair $\langle x, y\rangle$, which was only defined when x and y are sets. Thus x can only be in the domain of a function if x is a set. Now consider any g-inductive class A, and consider any element x in A. Then we know that $g(x) \in A$, and thus, because x is in the domain of g, x is a set. If it happens that A is finite (which could occur if, for instance, $\emptyset$ was a fixed point of g and A only contained $\emptyset$), A would be a set as a consequence of Exercise 6.1. Indeed, there are non-denumerable g-inductive classes, for instance, the class of all sets, which is g-inductive under the power function.

Section 5

Exercise 5.1

Suppose that the natural number n is in A and that for any number x in A, x^+ is also in A. We can prove that A contains every number $\geq n$ by g-induction. In particular, consider the property P: If $m \geq n$, then $m \in A$. We show that this prpoerty holds for all numbers m. Either $\emptyset = n$ or $\emptyset < n$. In the first case, P holds, while in the second case, P vacuously holds. Thus P holds for $\emptyset$. Now suppose that P holds for x. There are two cases: either $x^+ < n$, $x^+ = n$, or $x^+ > n$. In the first case, P vacuously holds. In the second case, because we assumed $n \in A$, P holds. In the last case, by the induction assumption that $x \in A$, $x^+ \in A$, and thus P holds for x^+. Thus, in any case, P holds for x^+, therefore completing the induction.

Exercise 5.2

We suppose f is a function from ω into A.

(a): We suppose that, for all n, $f(n) \subseteq f(n^+)$. We can prove that, for all n and m, if $n \leq m$ then $f(n) \subseteq f(m)$ by induction on m. In particular, we show that, for all numbers m, for all numbers $n \leq m$, $f(n) \subseteq f(m)$. The base case is when $m = \emptyset$. The only natural number $\leq \emptyset$ is $\emptyset$, which is a subset of itself, meaning the base case holds. Now suppose that the hypothesis holds for some m. Thus, for all $n \leq m$, $f(n) \subseteq f(m)$. Now, we are given that $f(m) \subseteq f(m^+)$. But this means that, for all number $n \leq m$, $f(n) \subseteq f(m^+)$. Thus all numbers $n \leq m^+$ are such that $f(n) \subseteq f(m^+)$, therefore completing the induction.

(b): We prove that, for all n, if $f(n) \subset f(n^+)$, then for all n and m, if $n < m$, then $f(n) < f(m)$. The base case is when $m = \emptyset$. In this case, there are no natural numbers less than $\emptyset$, so the property vacuously holds. No suppose it holds for some m. Thus, for all $n < m$, $f(n) \subset f(m)$. Now, we are given that $f(m) < f(m^+)$. But this means that, for all $n < m$, $f(n) \subset f(m^+)$, thus completing the induction.

Section 7

Exercise 7.1

We can prove the equivalent statement that, if a set S of natural numbers is not finite, then it is denumerable. Note that, because ω is the set of all natural numbers, $S \subseteq \omega$. We can make a 1-1 correspondence between S and ω as follows: pick some element a_0 of S and pair it with 0. Then pick another element a_1 of S and pair it with 1, and, in general, we pair n with some element a_n of S, where a_n is disitinct from all a_m where $m < n$. If there was any n for which it could not be paired to an a_n, this would mean that S only had $n-1$ elements, contrary to the assumption that it was infinite. If there was ever some a_n for which no n could be found, this would contradict the fact that $S \subseteq \omega$. Thus this really is a 1-1 correspondence.

We can also show that every non-denumerable set has a denumerable proper subset. For consider any non-denumerable set A. We can show that some proper subset of A can be put into a 1-1 correspondence with ω in the following way: Pick some element a_0 of A, and pair it with 0. Then pick another element a_1 of A, and pair it with 1, and in general, for every n, we pick a new element a_n of n to match with n. If for any n we could not pick a new element of A to match with it, that would mean that A would only have $n-1$ elements, contrary to the assumption that it was infinite. Thus, for every n, we can pick an element of A to match with it, meaning we have a denumerable subset S of A consisting of the elements $a_0, a_1, ..., a_n, ...$ If this was the entirety of A, then A itself would be denumerable, contrary to the assumption that it was not. Thus S is a proper subset of A, therefore completing the proof.

Having shown the above result, we can now prove that any infinite subset of a denumerable set is denumerable. For consider some denumerable set A. Suppose it had a non-denumerable subset B. By the above result, there is a proper subset C of B which is denumerable. Because $B \subseteq A$ and $C \subset B$, $C \subset A$. Thus we would have A being in a 1-1 correspondence with C, a proper subset of A, which is impossible. Hence A can have no non-denumerable subset.

Note the following incorrect proof that any denumerable set can be put into a 1-1 correspondence with any non-denumerable: Consider any denumerable set A and any non-denumerable set B. Arrange the elements of A in the order $a_0, a_1, ..., a_n, ...$ If for any n, a_n did not have a matching element in B, then B would be finite, a contradiction. Also, if we ever encountered an element in B for which we could pair no element of A with, this would mean A would be finite, a contradiction. The error in this proof is that we of course can reach an element in B for which we could pair no element of A with (and this does not mean that A is finite; it would only mean so if the statement was:"if we ever encountered some *nth element* in B for which we could pair no element of A with" This would then mean that A was finite). Of course, this element in B that we could not pair up with A would come after pairing off a denumerable amount of elements from B, meaning we would already know that A was infinite (and thus there is no supposed contradiction).